

Version Control Is a Job Skill, Not Just a Tool

Thinking about version control the way professional game developers do

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Everyone can probably relate to this

Overwritten by a teammate. Lost to a laptop crash. Or tangled in a naming scheme that stopped making sense three files ago. This isn't unique to games, but it hits us extra hard because of how interconnected all the parts of a game are.



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WHY THIS MATTERS

This is already table stakes.

94%

of developers surveyed in 2026 now use version control, up from 86% a year earlier.

Build like the pros from day one and you never have to retrofit it later, when it hurts.

37%

of game-industry respondents say moving large files is slow or difficult. Across all industries, it is the most-cited collaboration problem.

Even for a small Godot project, the art assets can add up, and Dropbox or Google drive makes performance and conflicts worse.

94%: Perforce 2026 State of Real-Time Workflows Report, all-industry. 37%: Perforce 2025 State of Game Technology Report, gaming respondents; 36% across all industries.

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Version control isn't corporate process. It's infrastructure for anything you want to survive past today.

Whether that's a portfolio piece, a passion project, or the start of your own studio.

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WHAT'S CHANGING

Programmer jobs fell. Developer jobs held steady.

−27.5%

US "programmer" employment, 2023–2025.

−0.3%

US "software developer" employment, same period.

In the government data, "software developer" is the more design-oriented role. Career advisers say these jobs also ask for knowledge of the whole development life cycle and for working with other people. Version control supports all of it, at a studio or on your own projects.

IEEE Spectrum, December 2025, citing U.S. Bureau of Labor Statistics data.

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LEVELLING UP

Learning the commands is the easy part. The decisions are the skill.

- How to structure a project so other people can work in it
- Which files to lock
- When to branch, and when to stay on main
- What belongs in version control at all
- How to get back when something goes wrong

In your own projects

Fewer hours of lost work, and a project that can grow beyond one person, whether that’s one more teammate or a whole studio.

In a job

Your portfolio might land the first interview. In later rounds, studios want to know whether you understand how professional teams work together, and whether you can join their production and contribute fast.

These are the decisions that take you from beginner to intermediate, then advanced, and on to architect.

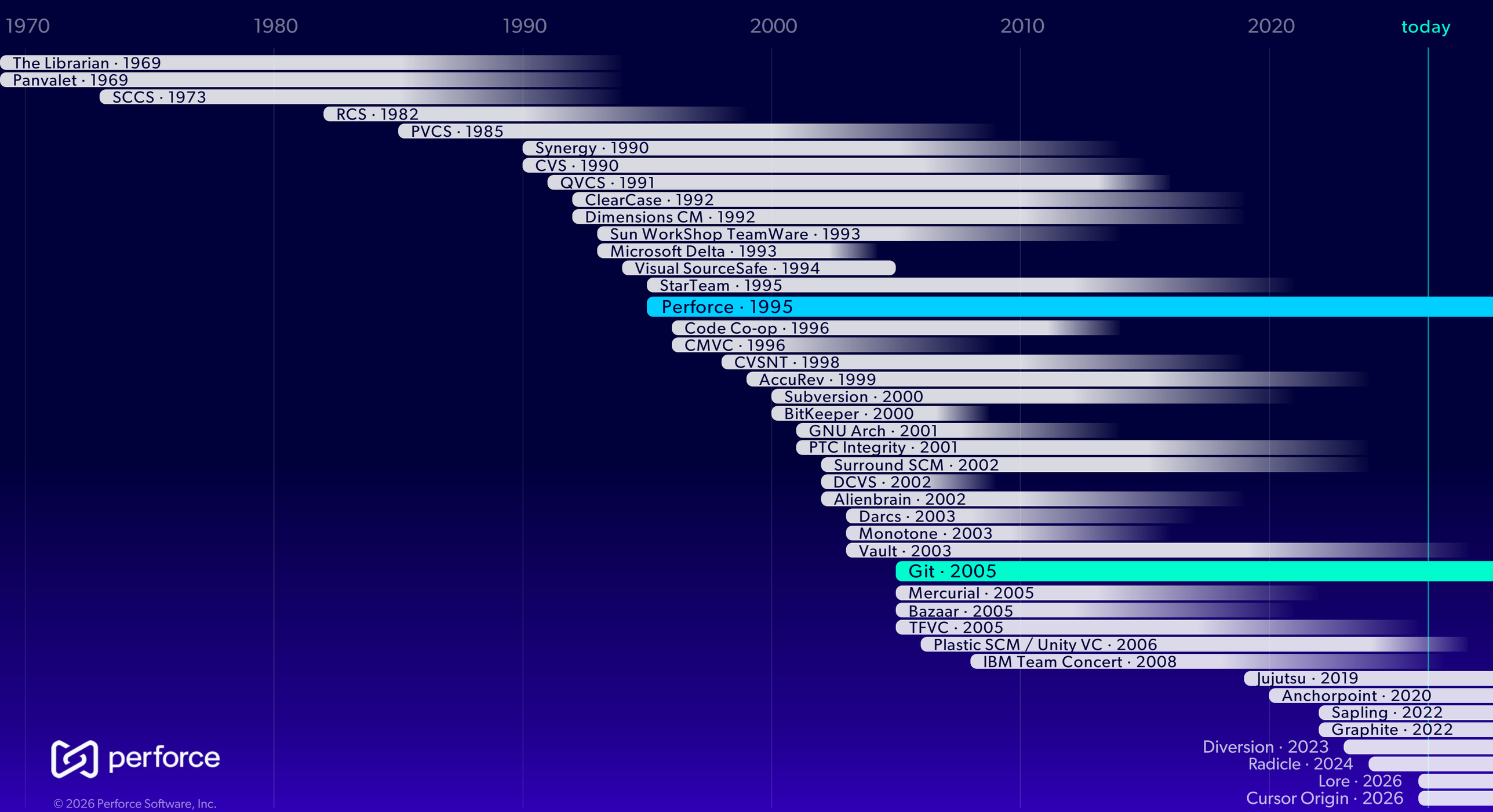
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A Brief History of Version Control

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Each one was built for a different problem.

Ten years apart, by two people with different constraints.

Christopher Seiwald, 1995

Written in a basement in Alameda, with the first servers in a paint closet.

The problem: a team in one building sharing one large, growing depot, on 1995 hardware. Everyone needed the same answer about what is current, right now.

So: one authoritative server, workspaces instead of copies, and speed as the headline feature. History is stored on the server, and ordinary users can't rewrite it. Microsoft licensed it in 1999 to build Windows rather than use its own tool.

Linus Torvalds, 2005

Written in about ten days, after BitKeeper's licence was pulled from the Linux kernel.

The problem: Contributors to Linux each needed their own full copy of history, frequent branching and merging, needed a way to ensure changes to the revision history can be verified since there is no central server.

So: every clone holds the whole history, branching and merging are the everyday operations, each repo can have its own separate history and history can be amended and changed.

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Git is not GitHub.

Git is open-source software. The others are products built around it.

Git

Open-source software, GPLv2, 2005. Anyone is free to run it, modify it, host it or build on it. That is part of why many new version control systems are built on top of it.

It defines commits, branches and merges — and nothing about servers, accounts or reviews.

A forge

GitHub, Bitbucket and Azure Repos are commercial services. GitLab ships a community edition; Gitea and Forgejo are open source.

Each adds the central network: a canonical remote, accounts, permissions, pull requests, branch protection. You can host your own.

And P4

A P4 server can also host Git repositories.

Code in Git, assets in the depot, one server behind both.

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THE BIG IDEA

Version control keeps the history, so you don’t have to.

Every change is recorded for you. There’s never a question of which version is the latest, and you can get back to any point in the past.

BY HAND

level_boss.psd
level_boss_final.psd
level_boss_final_v2.psd
level_boss_FINAL_real.psd
project_backup_0312/

Which one is the latest, and who changed what? Nobody is sure.

WITH VERSION CONTROL

level_boss.psd
#4 Fix boss lighting latest
#3 Add arena props
#2 Block out arena
#1 First pass

One file. The system records every version, and any of them is one step away.

FOUR CONCEPTS WE’LL COVER

1 The repository
2 Branches
3 Atomic changes
4 Where changes live

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TWO VOCABULARIES, ONE SET OF IDEAS

Most version control concepts are shared

THE IDEA	IN GIT	IN P4
Where the history is kept	repository	depot
Your local copy of the files	clone / working tree	workspace
A set of changes, recorded together	commit	changelist
Getting everyone else's changes	pull	sync
Publishing yours	push	submit
Setting aside changes you don't want to submit yet*	stash	shelve*
A separate line of development	branch	stream or branch
How you request a review before submitting	branch/fork + pull request*	shelve + review

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CONCEPT 1

The repository is the whole story.

Git calls it a repository, P4 calls it a depot. Same job: it keeps every version of every file, along with extra data about who, why, what, and when.

CHANGE	DESCRIPTION	USER	TIME
#1482	Add hairball projectile VFX VFX/fur_trail.tres #1 add Art/FX/collar_diffuse.png #1 add Scripts/laser_pointer_follow.cs #6 edit	mira	10 min ago
#1481	Fix double jump off catnip Scripts/Player/Controller.cs #12 edit	sam	this morning
#1480	Scratching posts in alley Levels/living_room.tscn #9 edit Props/scratching_post.tscn #1 add	theo	yesterday
#1479	New Whiskers idle loop Art/Characters/whiskers_idle.fbx #4 edit	mira	yesterday

…and every version before that, back to the first one.

Save points at every change

You can open any file as it was at any point, and restore it.

Who, when and why

Every submission has an author and a message. Good descriptions give context to the changes.

Single Source of Truth

No separate folders on a laptop, not a zip file in a chat thread, no debate about what is “latest.”

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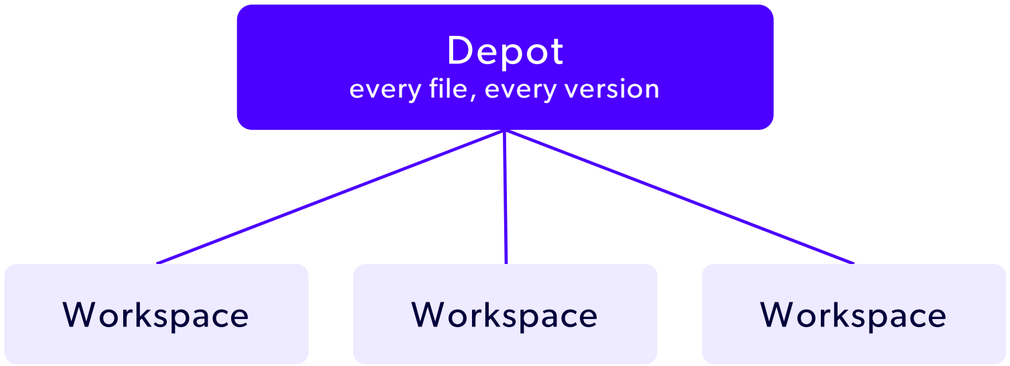
CONCEPT 1

The repository is the single source of truth.

Git calls it a repository, P4 calls it a depot. It stores every version of every file and its full history. The two philosophies differ on where it lives.

CENTRALIZED

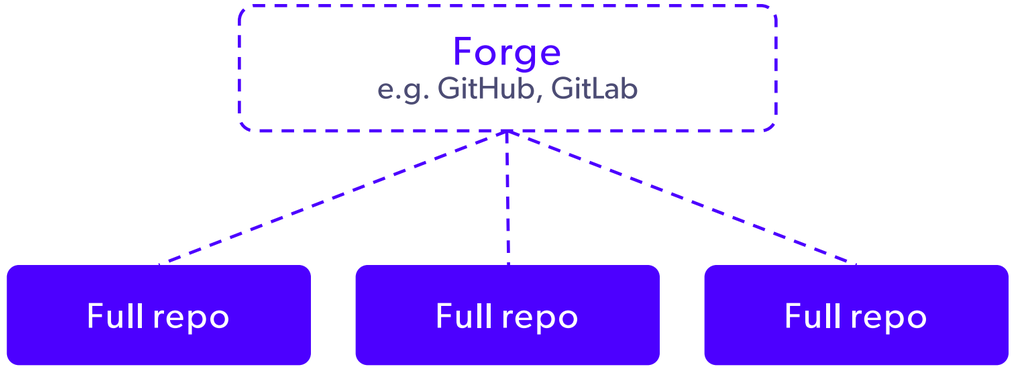
One central depot



Everyone submits to and syncs from one server. Workspaces hold only the files each person needs. P4 works this way.

DISTRIBUTED

A full repository on every machine



Each clone holds the full history and works offline. A forge keeps the copy the team treats as the source of truth. Git works this way.

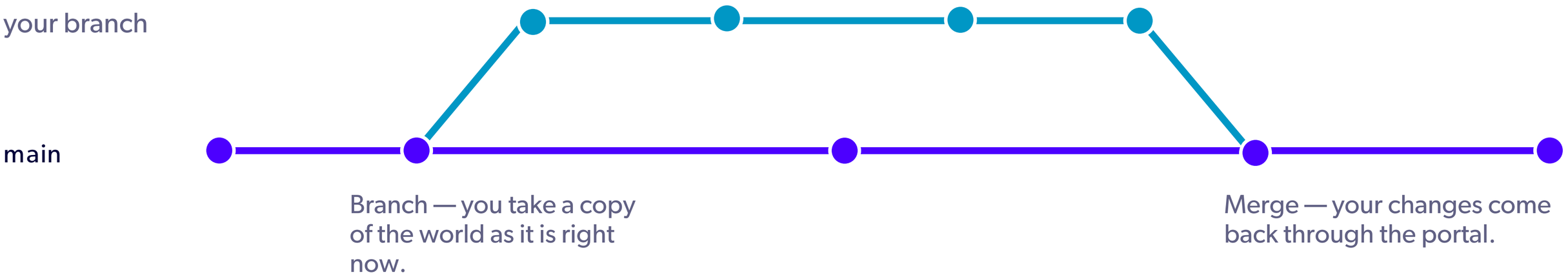
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CONCEPT 2

A branch is a parallel universe.

You step sideways into your own copy of the project, work freely, and bring it back when it's ready.



Main keeps working the whole time you're away. And if the idea turns out badly, you throw the universe away instead of trying to pull the changes out.

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CONCEPT 3

Atomic Changes: All or Nothing

"Atomic" mean the whole set lands together, or none of it does.

Changelist #147 — boss arena lighting

- arena_boss.mesh
- arena_boss.material
- BossLights.cs

SUBMIT

CHANGE	DESCRIPTION	USER
#147	Boss arena lighting	mira
#146	Boss phase 2 timing	sam
#145	Arena floor textures	theo
#144	Boss intro cutscene	mira

…and every change before it, back to the first one.

A mesh without its material is a broken game. Nobody ever syncs two files out of three.

And every point in the history is a specific, intentional snapshot in time — so "go back to #146" is a complete change, less hunting for related files.

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CONCEPT 4

Changes to different lines merge on their own.

You and Sam both edited boss_ai.cs, but each of you changed a different line.

YOUR VERSION

```
1  class BossAI {
2    float speed  = 6.0f;
3    int   health = 500;
4    float range  = 12.0f;
5    int   damage = 25;
6  }
```

You changed line 2: speed.

+

SAM'S VERSION

```
1  class BossAI {
2    float speed  = 4.0f;
3    int   health = 500;
4    float range  = 12.0f;
5    int   damage = 40;
6  }
```

Sam changed line 5: damage.

=

MERGED RESULT

```
1  class BossAI {
2    float speed  = 6.0f;
3    int   health = 500;
4    float range  = 12.0f;
5    int   damage = 40;
6  }
```

Both kept, merged automatically.

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CONCEPT 4

Same-line changes need a person to decide.

You and Sam both changed the boss's health. The tool can't guess which one wins.

YOUR VERSION

```
2    float speed  = 4.0f;
3    int   health = 800;
4    float range  = 12.0f;
```

SAM'S VERSION

```
2    float speed  = 4.0f;
3    int   health = 650;
4    float range  = 12.0f;
```

→

CONFLICT IN BOSS_AI.CS

```
2    float speed  = 4.0f;
     <<<<<<< yours
3    int   health = 800;
     =======
3    int   health = 650;
     >>>>>>> sam
4    float range  = 12.0f;
```

The tool marks both versions and stops.

→

YOU RESOLVE IT

```
2    float speed  = 4.0f;
3    int   health = 800;
4    float range  = 12.0f;
```

Keep yours, keep Sam's, or write a mix. Then submit.

Binary files can't merge at all. Art, audio and scenes have no lines to compare, so lock them before you edit.

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SAME IDEA, DIFFERENT MECHANICS

A few key differences between Git and P4

Saving versus sharing

In **Git** it's two steps:
commit records locally, push shares it.
In **P4** a submit does both at once, straight to the history everyone shares on the server.

Unfinished work

A **Git** stash is yours alone.
A **P4** shelf lives on the server, so a teammate can unshelve it, review it, or finish it for you.
(This is also what is used for Code Reviews instead of a forge's Pull Request)

Permissions

In **Git** if you have access to clone the repo you get everything in it.
P4 permissions can be set for specific paths within a depot, so certain groups can be allows to read or write different files.

What you get locally

A **Git** clone brings the whole repository and its whole history.
P4 workspace syncs only the specific versions of the file paths you ask for — useful when the project is hundreds of gigabytes.

Branching

Git branches are local so they don't exist for anyone else until you push them. They are fast to create and can be merged with any other branches (no rules).
P4 branches and streams are defined on the server for everyone. Streams are set up with rules about what merges where and which should be more stable.

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TWO WORKFLOWS

Branch for every change, or share one stream.

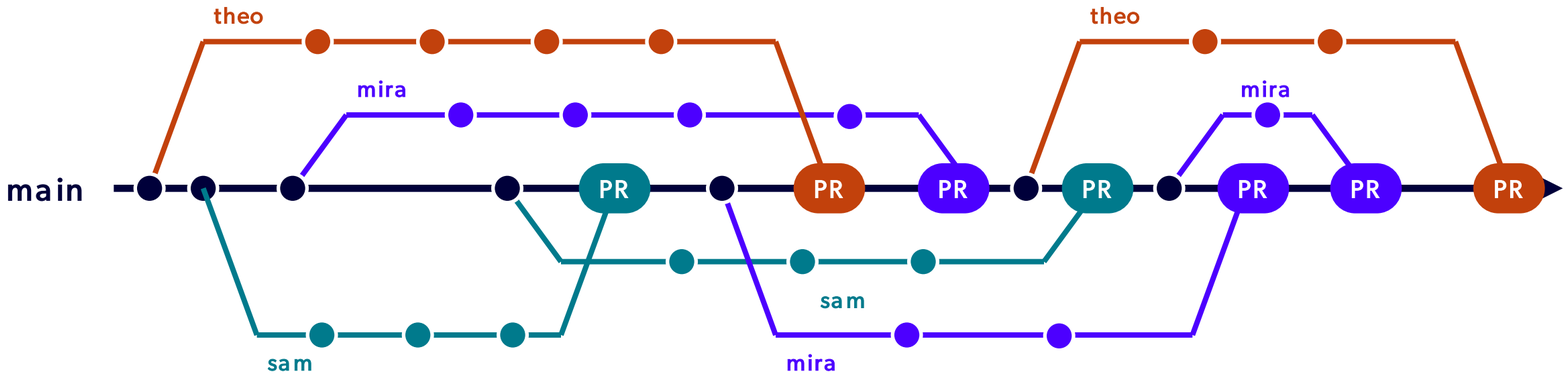
Branching strategies is a whole field on its own, but these are the most common ways it is done in Git vs P4

GIT-SHAPED

A branch for every change

Everyone branches for each fix, so many branches are open at once.

A pull request reviews each one and merges it into main.

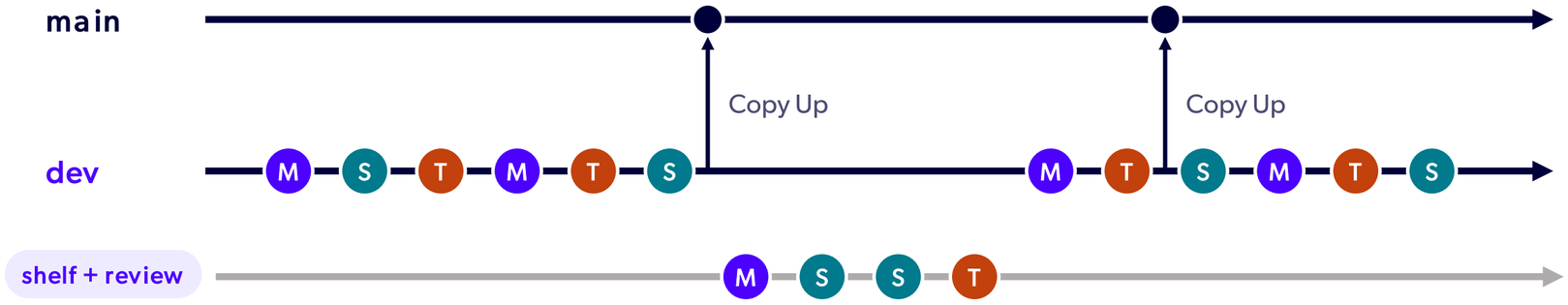


P4-SHAPED

One stream for everyone

Mira, Sam and Theo all submit to one dev stream, with shelves for review.

One shared stream keeps file locking simple.



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SAVING YOUR WORK

Git stages after you edit. P4 checks out before.

The order decides when the team finds out what you are working on.

GIT-SHAPED

Edit first, then stage

Change any file at any time. Git notices what changed.

Staging picks which changes go into the commit.

1
Edit
Change any files you like

→

2
Stage
`git add`
Pick changes to commit

→

3
Commit
`git commit`
Saved in your local repo only

→

4
Push
`git push`
Shared with the team

P4-SHAPED

Check out first, then edit

Files stay read-only until you check them out.

The team sees who has each file, and binaries can lock.

1
Check out
`p4 edit`
Team sees who has it; binaries can lock

→

2
Edit
Change the files you checked out

→

3
Submit
`p4 submit`
Saved and shared in one step

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Git CLI

```
sam@git $  git pull
From https://github.com/team/PurrForce
   65779e4..bafcff7  main       -> origin/main
Updating 65779e4..bafcff7
Fast-forward
 levels/living_room.tscn | 6 +++---
 1 file changed, 3 insertions(+), 3 deletions(-)
```

Open file **cat/cat.gd** in Godot/VS Code/etc. and edit it.

```
sam@git $  git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   cat/cat.gd

no changes added to commit (use "git add" and/or "git commit -a")

sam@git $  git add cat/cat.gd

sam@git $  git commit -m "Add pounce"
[main 6c7e1e2] Add pounce
 1 file changed, 3 insertions(+)

sam@git $  git push
To https://github.com/team/PurrForce.git
   bafcff7..6c7e1e2  main -> main
```









P4 CLI

```
sam@p4 $  p4 sync
//purrforce/main/levels/living_room.tscn#2 - updating
C:\PurrForce\levels\living_room.tscn


sam@p4 $  p4 edit cat/cat.gd
//purrforce/main/cat/cat.gd#1 - opened for edit
```

Open file **cat/cat.gd** in Godot/VS Code/etc. and edit it.

```
sam@p4 $  p4 status
cat\cat.gd - submit change default to edit //purrforce/main/cat/cat.gd#1




sam@p4 $  p4 submit -d "Add pounce"
Submitting change 3.
Locking 1 files ...
edit //purrforce/main/cat/cat.gd#2
Change 3 submitted.
```

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Specialized Version Control Features

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Large File Handling

Git size limits

Git was designed for text files originally, so it has a small file size limit natively.

- When a repository gets large it becomes very slow
- Forges usually have a size limit per file, too
- Git LFS is an addon that can be added to git
  - LFS is configured to store certain file types in a separate storage location, usually provided by your forge (e.g. github, gitlab,etc)
- Your forge will also have specific storage limits.

```
sam@git $  git push
remote: error: File art/cat_fur.psd is 123.00 MB; this exceeds GitHub's file size limit of 100.00 MB
remote: error: GH001: Large files detected. You may want to try Git Large File Storage - https://git-lfs.github.com.
 ! [remote rejected] main -> main (pre-receive hook declined)
```

```
sam@git $  git push
batch response: This repository exceeded its LFS budget. The account responsible
for the budget should increase it to restore access.
```

Example, GitHub Free tier:
100 MiB per file in Git
2 GB per file in LFS
10 GiB of LFS storage (total of all versions of all files)

Perforce P4 was built for binary and text from the start, which is why it is most commonly used in gaming, VFX, animation, and other industries that use images, audio, 3D models, CAD files, etc.

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Exclusive Checkout

Built-In File Locking

What happens if two people make changes to the same image, model or other binary file?

In Git

Because git is naturally offline-first, a user doesn’t know what others are working on.
When the second user tries to push, they get an error.
Nothing can merge two PNGs, so one version is chosen and the other person's work has to be deleted and redone.

In Perforce

Specific file paths can be configured as *exclusive-checkout* on the server
The first check-out locks the file.
The second person is stopped from checking it out until the first user either reverts it back or submits a new version.
This prevents the conflict, *before* you wasted hours of work

This is the most common example of distributed vs centralized version control systems

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Jase's Hot Tip

Automate your builds *before* you even start

Just setting up a manual script you can run from any machine is better than nothing

A classic way a team misses a deadline: the build broke hours or days ago and nobody noticed until it's become a big job to fix.

```
p4 sync
build.bat
smoke_test.exe
# does it launch?
# does the main scene load?
```

A quick, short build script is all it takes.

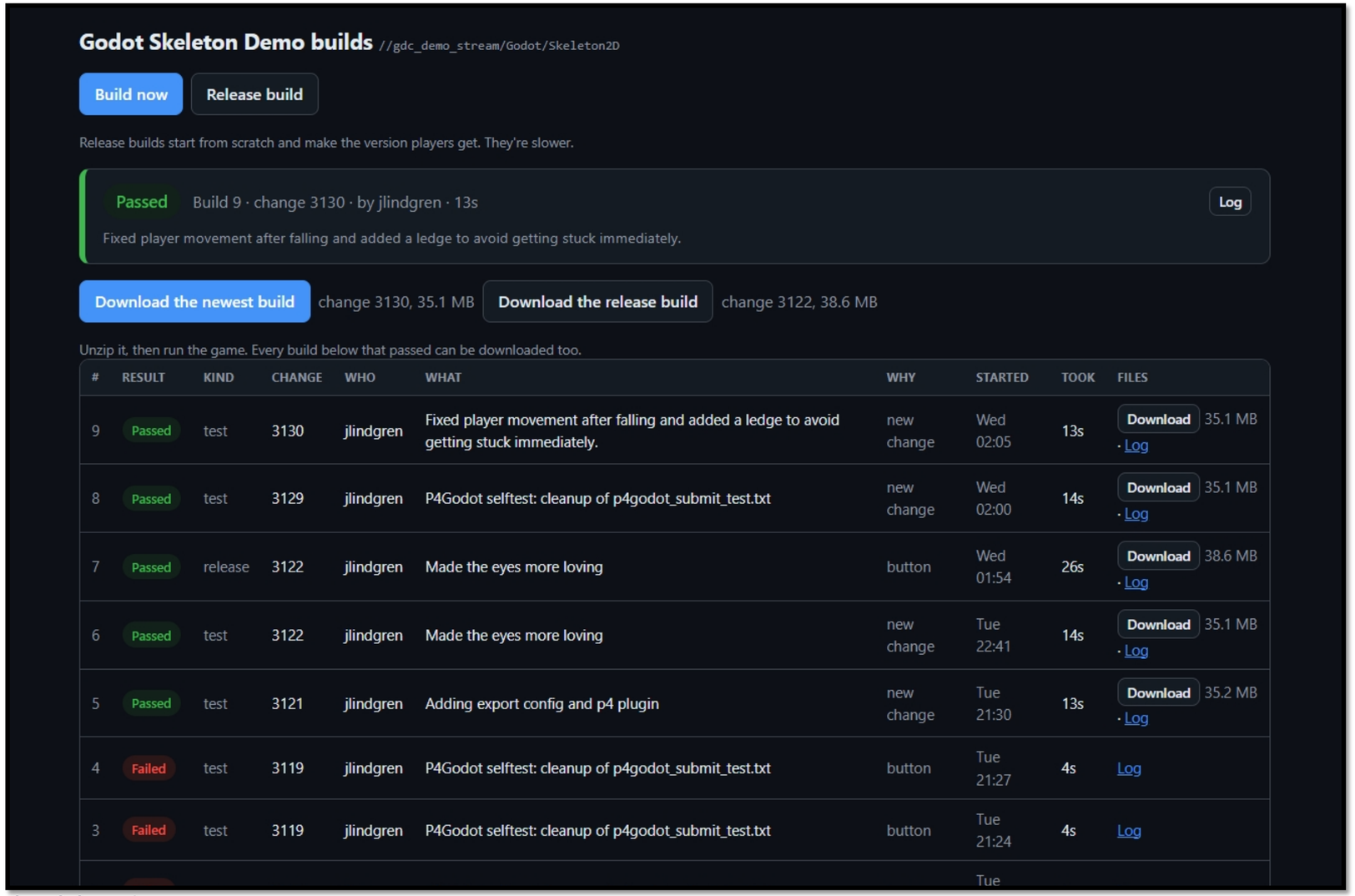
You can grow into Jenkins, TeamCity or Epic's Horde when you're ready.
You don't need any of that today — you just did it by hand.

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Jase's Hot Tip

Set this up *before* your next project



Godot Skeleton Demo builds //gdc_demo_stream/Godot/Skeleton2D

Build now | Release build

Release builds start from scratch and make the version players get. They're slower.

Passed Build 9 · change 3130 · by jlindgren · 13s — Log

Fixed player movement after falling and added a ledge to avoid getting stuck immediately.

Download the newest build change 3130, 35.1 MB | Download the release build change 3122, 38.6 MB

Unzip it, then run the game. Every build below that passed can be downloaded too.

#	RESULT	KIND	CHANGE	WHO	WHAT	WHY	STARTED	TOOK	FILES
9	Passed	test	3130	jlindgren	Fixed player movement after falling and added a ledge to avoid getting stuck immediately.	new change	Wed 02:05	13s	Download 35.1 MB · Log
8	Passed	test	3129	jlindgren	P4Godot selftest: cleanup of p4godot_submit_test.txt	new change	Wed 02:00	14s	Download 35.1 MB · Log
7	Passed	release	3122	jlindgren	Made the eyes more loving	button	Wed 01:54	26s	Download 38.6 MB · Log
6	Passed	test	3122	jlindgren	Made the eyes more loving	new change	Tue 22:41	14s	Download 35.1 MB · Log
5	Passed	test	3121	jlindgren	Adding export config and p4 plugin	new change	Tue 21:30	13s	Download 35.2 MB · Log
4	Failed	test	3119	jlindgren	P4Godot selftest: cleanup of p4godot_submit_test.txt	button	Tue 21:27	4s	Log
3	Failed	test	3119	jlindgren	P4Godot selftest: cleanup of p4godot_submit_test.txt	button	Tue 21:24	4s	Log
							Tue		

Free and open-source on Github
jase-perf/p4-build-machine-lite

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AT A STUDIO

The same idea, at studio scale.

The build check you just saw is what studio build systems do for hundreds of people: Jenkins, TeamCity and Epic's Horde, connected to a sync tool like UnrealGameSync that shows which change is safe to sync. Knowing what these do, and that version control supports all of them, tells you what to grow your own pipeline into, and it's something you can talk about in an interview.

41% / 23%

of AAA studios use Jenkins / TeamCity for automated builds and tests (CI/CD), the two most common.

31%

of indie studios said they weren't using CI/CD tools.

Perforce 2024 State of Game Technology Report, with JetBrains.

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THE AI ERA

Where AI does the task, young workers fell behind. Where AI helps the person, jobs held steady.

19% how far employment of 22–25-year-olds in highly AI-exposed jobs has fallen behind the trend for the same age group in less-exposed jobs (June 2026), mostly through less hiring.

In jobs where AI is used to help people rather than replace their tasks, employment is flat or rising, especially for experienced workers. Employment fell for young workers in jobs built on knowledge you can learn from a textbook, and rose for experienced workers in jobs built on knowledge you get through practice, mentorship and real situations.

Version control judgment is the second kind. You build it on real projects, in a studio or on your own.

Stanford Digital Economy Lab, “Canaries in the Coal Mine?”, revised August 2026. ADP payroll data, US.

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THE AI ERA

AI can run the commands. The decisions are still yours.

AI coding tools can already commit, branch and resolve simple conflicts. What you still have to decide: how a project should be structured so a whole team can work in it, which assets lock, and what branching looks like at release. And when an AI tool makes a bad version-control decision, someone has to notice and get the project back.

Knowing the fundamentals is what lets you direct the tools instead of fixing their mistakes.

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Getting Started Is Free

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WHAT IT COSTS

Free to get started

P4

Free for up to 5 users and 20 workspaces. Full-featured — the same server AAA studios run.

P4 ONE

The newer client, with a simpler interface that bridges the centralized and local models.

Free for solo use, works with the free server.

Git

Free and open source, with GitHub, GitLab and Bitbucket free for personal use.

Learn it too. You'll need both.

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AN HOUR, ON THE MACHINE YOU HAVE

Try it yourself, start to finish.

1

Stand up your own server

You can install on P4 Cloud or just self-host for free on your own computer to get started.

2

Configure it for a game project

Set file types so art and audio lock, then create a depot and a stream. See the P4 Quickstart for Examples.

3

Create a workspace

Your workspace is the folder where you make changes, before you submit them back to the server.

4

Submit, and keep building

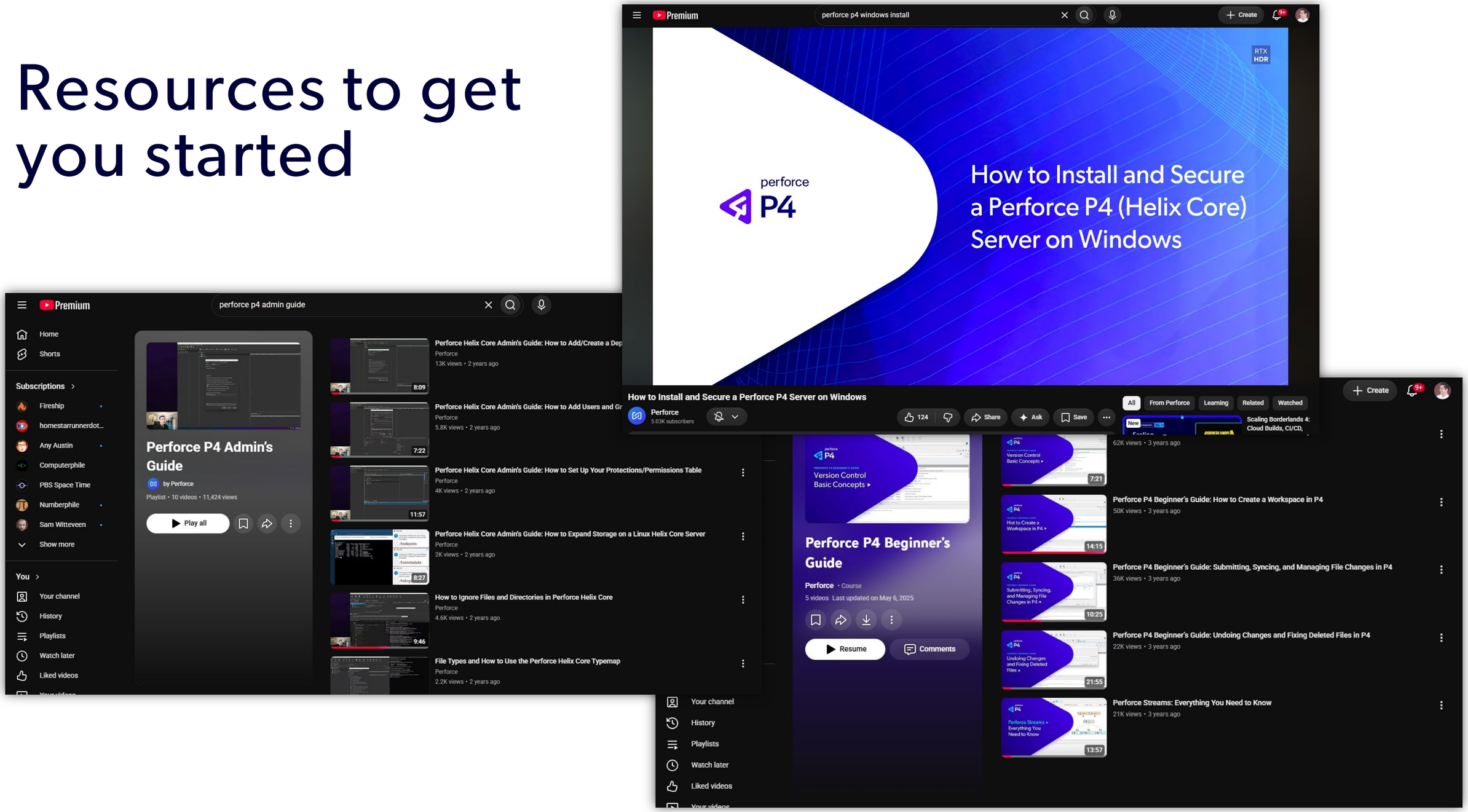
Submit the project, make sure builds are working, then keep going! Add team members and make sure to back up your server.

Do this when nothing is on fire. The point is to learn early while the mistakes cost less.

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Resources to get you started



Resources From This Talk



jase.town/igda-2026

Free tools

Open-source tools to enhance P4

Links to Documentation

Official downloads and Quickstart Guides

P4 Build Machine Lite

A tool to jumpstart your automated builds

P4Godot Plugin

Brand new open-source plugin for Godot

Reference p4ignore files and typemaps

Make setting up your server easier

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Thank You

Jase Lindgren

Principal User Advocate, P4

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